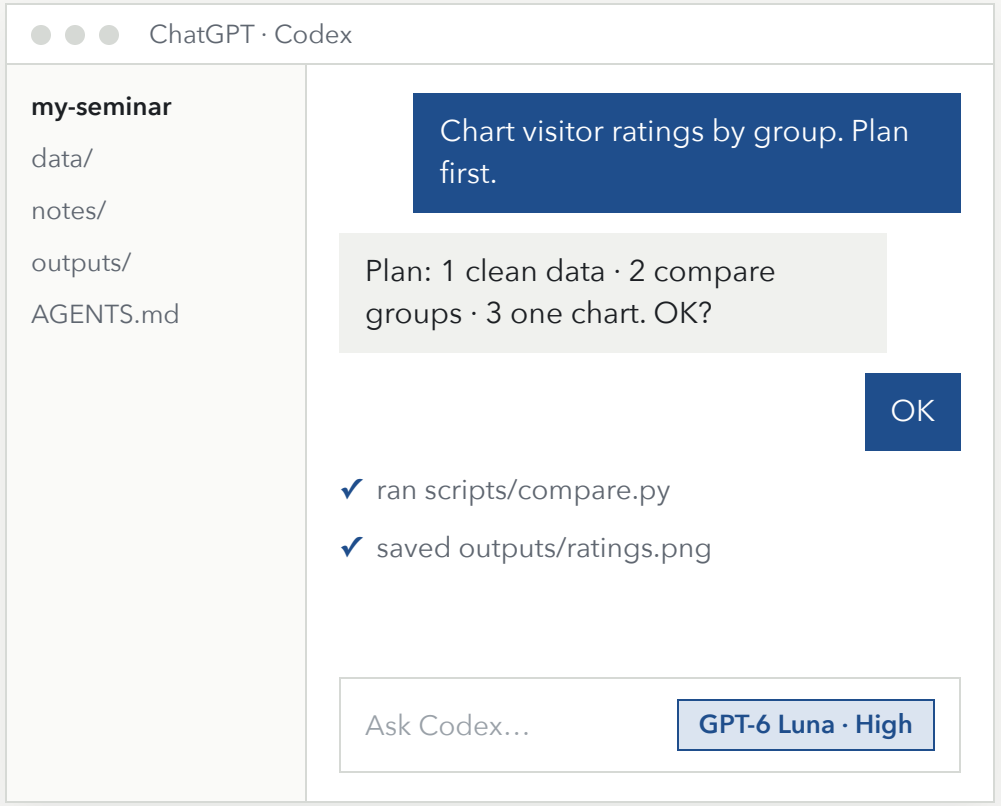
Seminar workshop

Codex for Research

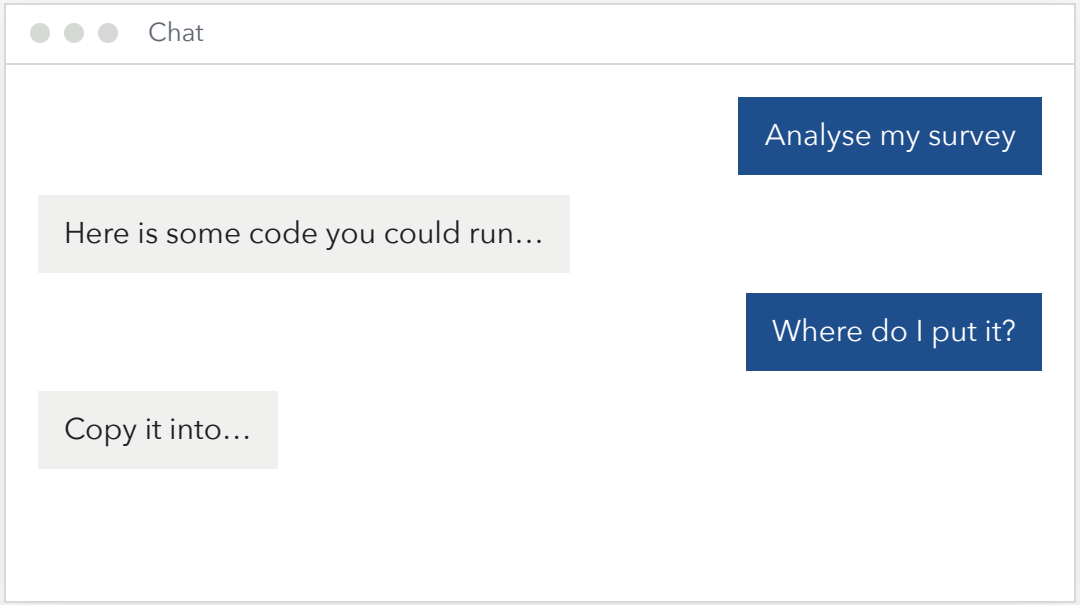
Let the agent do the work. Keep the judgement.

Pann Phetra, College of Sustainability and Tourism
Ritsumeikan Asia Pacific University



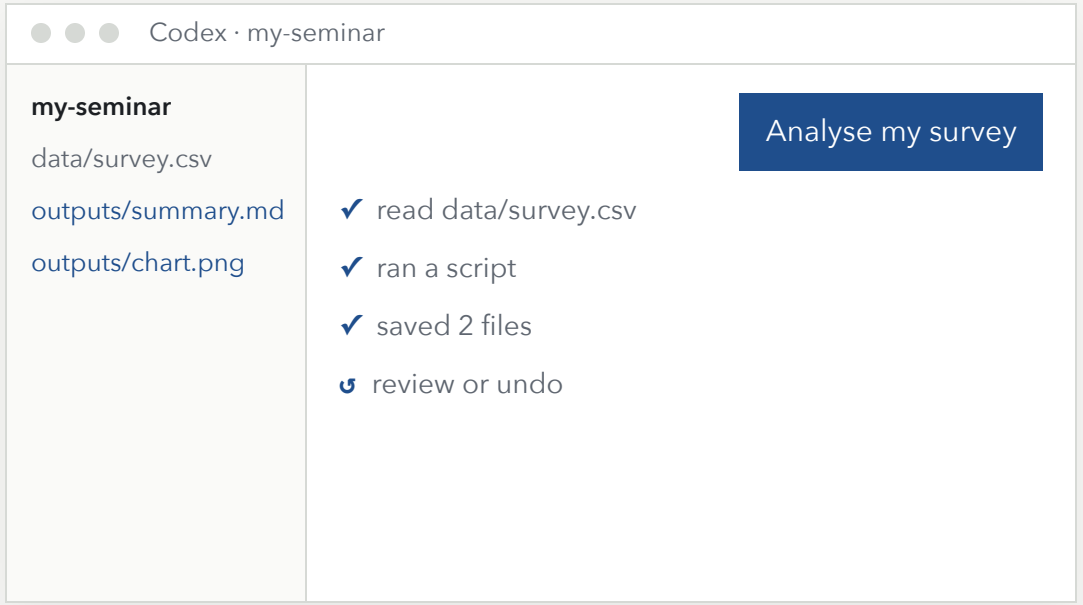
A chatbot answers. An agent works in your folder.

Chatbot



You copy things in and out by hand

Agent (Codex)



It reads, writes and runs files, and you review every change

What you need

1

ChatGPT account

Free · GPT-6 Luna

+

2

Desktop app

macOS · Windows · Linux

+

3

One folder

```
my-seminar/
  data/  notes/
```

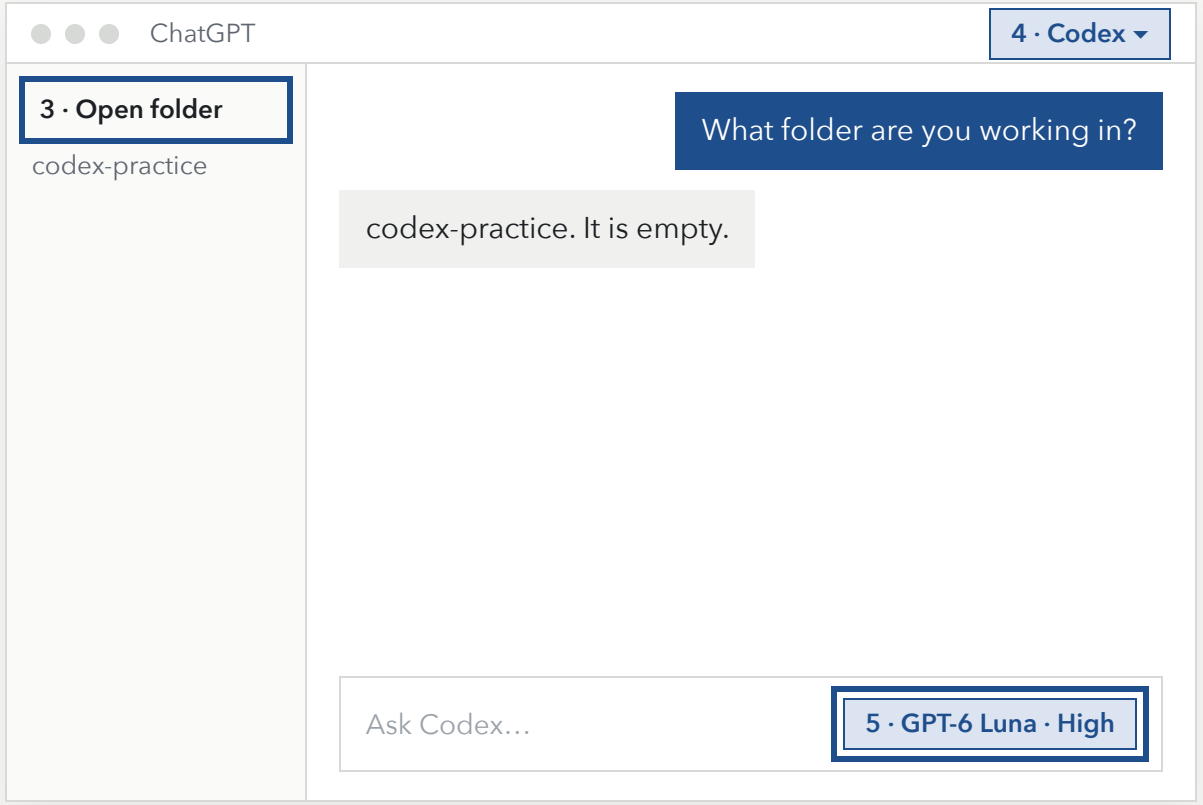
> Guide: github.com/Pann13223029/codex-seminar

Get it: **Code** → **Download ZIP**, or paste the link into Codex and ask it to download and guide you

Install in six steps

1. **Download** the ChatGPT app
2. **Sign in**
3. **Open a folder**
4. **Choose Codex**
5. **Check the model**: Luna High
6. **Check it works**

learn.chatgpt.com/docs/quickstart

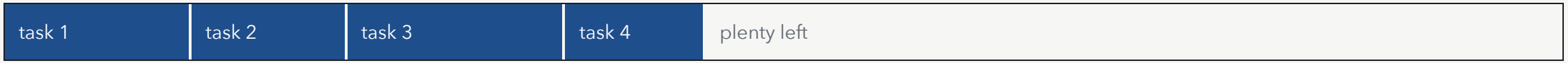


On the Free plan, make your usage last

Your allowance: one long, messy chat



The same allowance: small, planned tasks



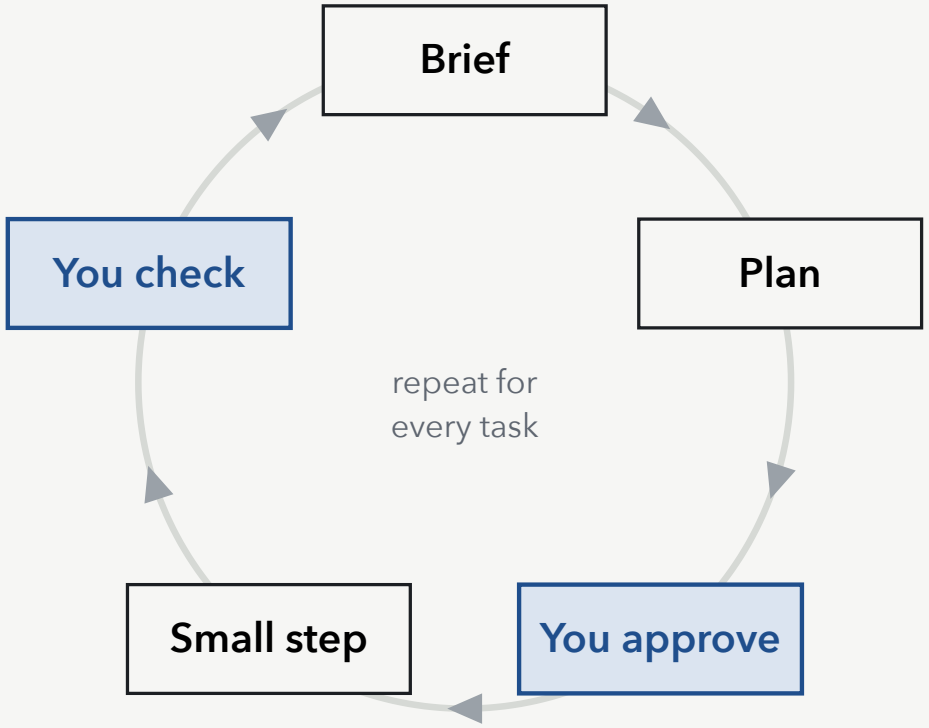
New chat per task

Light effort for quick jobs

Plan before work

Draft prompts elsewhere

The way of working



Codex: brief, plan, step

It reads the brief, proposes, then does one step

You: approve and check

The two blue boxes are yours. Never skip them.

AGENTS.md: a brief Codex reads before every task

AGENTS.md

```
# Project
Question: what do visitors rate lowest?

## Files
data/survey.csv

## Rules
- Plan first; wait for my OK
- Never change the original data
- Never invent numbers or quotes
```



- Task 1: describe the data
- Task 2: compare two groups
- Task 3: make one chart

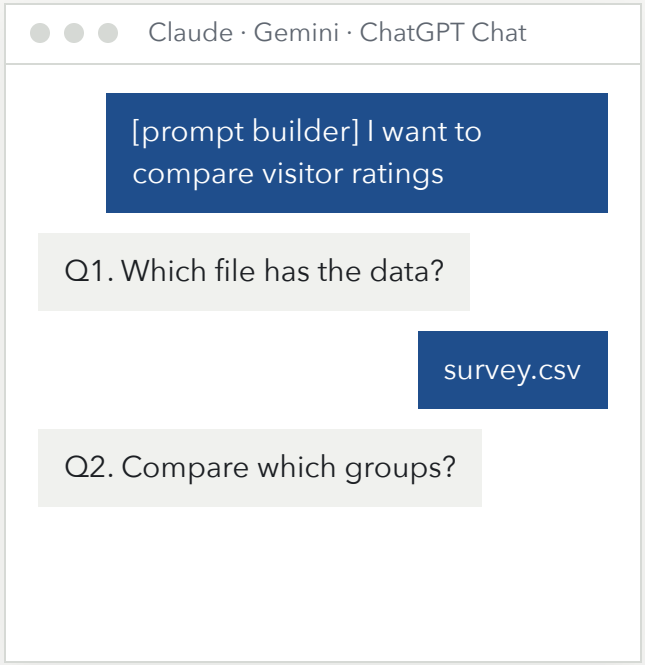
Write it once. Your rules apply to every task. Codex can write it for you (step 02).

A good prompt has six parts

1 Goal	Do international visitors rate Beppu lower? For my talk.
2 Context	data/survey.csv. "International" = not from Japan.
3 Task	Describe the data, then compare the two groups.
4 Output	A table in outputs/results.csv and one bar chart.
5 Checks	Give me one number to check by hand.
6 Limits	Do not change the file. Plan first, wait for my OK.

Not "analyse my data". Examples to copy: prompt-guide/

Let a regular chatbot write the prompt



→

Six-part prompt

```
Goal: …
Context: [file]
Task: …
Output: …
Checks: …
Limits: …
```

→

Codex

you paste it in

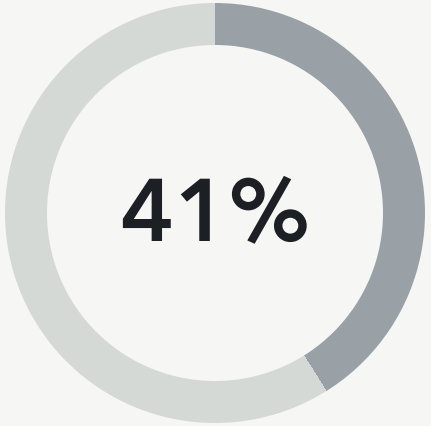
Saves your Codex usage

Fill in the [placeholders] yourself

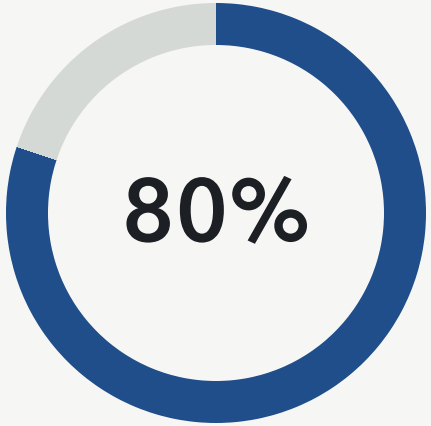
Never paste private data

Example 1: my graduation thesis

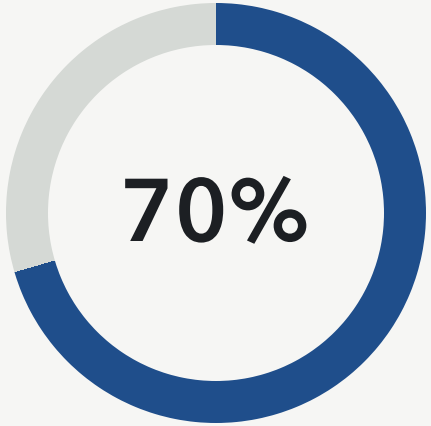
"4 in 10 incinerators generate electricity." Is that low?



of plants



of waste



of capacity

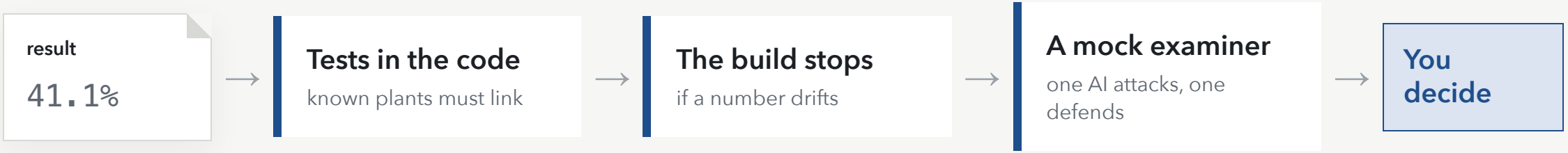
Big plants generate. 20 years of Ministry data, 23,593 records linked into 1,690 plant histories. FY2024.

Who did what

	Data	Analysis	Writing	Meaning
AI	code to read 20 spreadsheets	runs the statistics I chose	figures, drafts	
Me	**what counts as one plant**	**definitions, checks by hand**	**final text, edits**	**the question and the answer**

> Fast hands, not the brain. The thesis declares how AI was used.

Make mistakes hard to miss

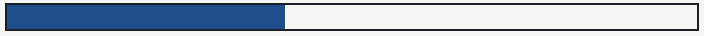


> Ask the agent to check its work, then check the check yourself.

Example 2: a data analytics capstone

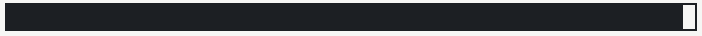
Google Merchandise Store, 903,653 web sessions. Which channels deserve the budget?

41%



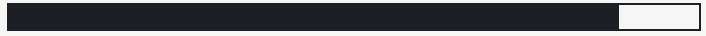
of revenue came from Google's own staff

98.8%



of a teaching model's top "buyers" were staff

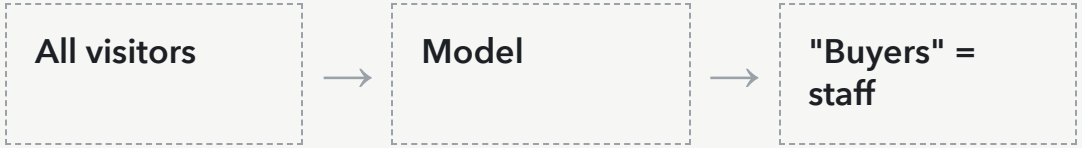
89%



of Display's credit was one buyer

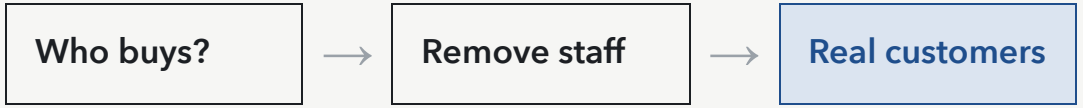
Ask who is buying before you model

Model first



✕ Looks accurate, targets the wrong people

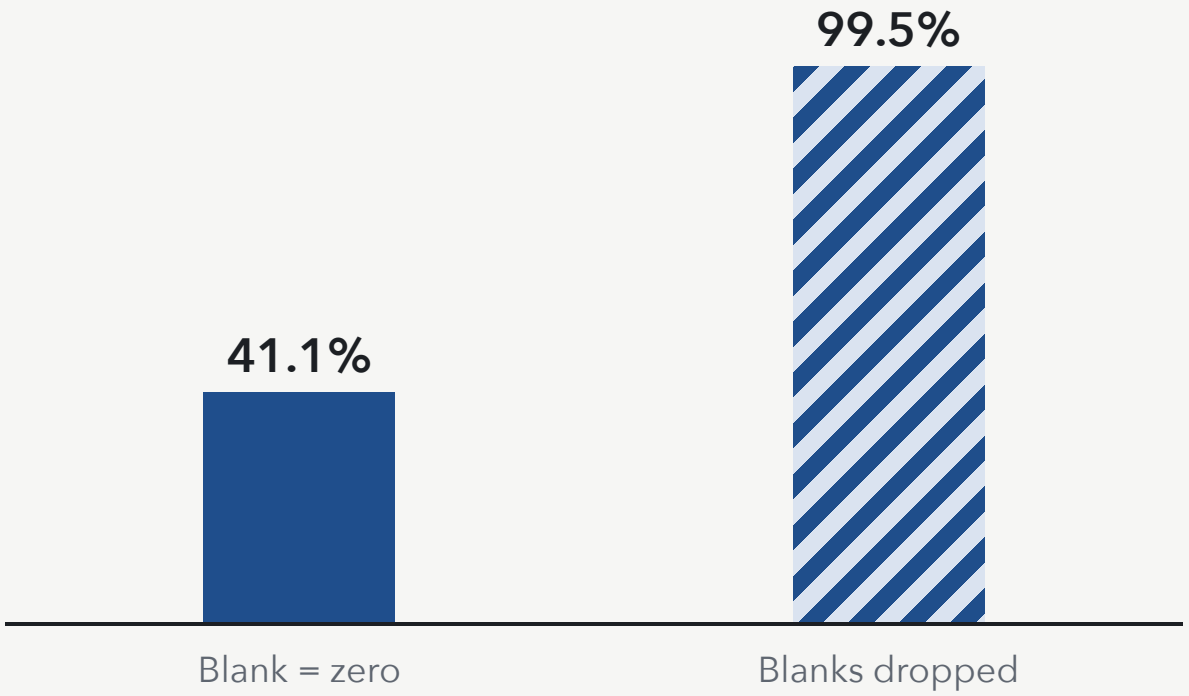
Question first



✓ Filter checked against the true source: 99.2%

> The AI stated facts from memory, twice wrongly.
> Checking the source caught it.

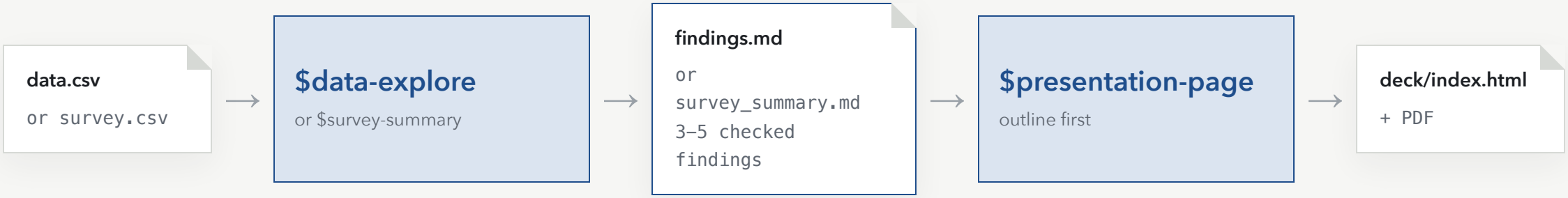
Two rules for blanks, two different answers



power_capacity_kw
1200
(blank)
0
(blank)

Always ask: how were missing values treated?

Three example skills, ready to use

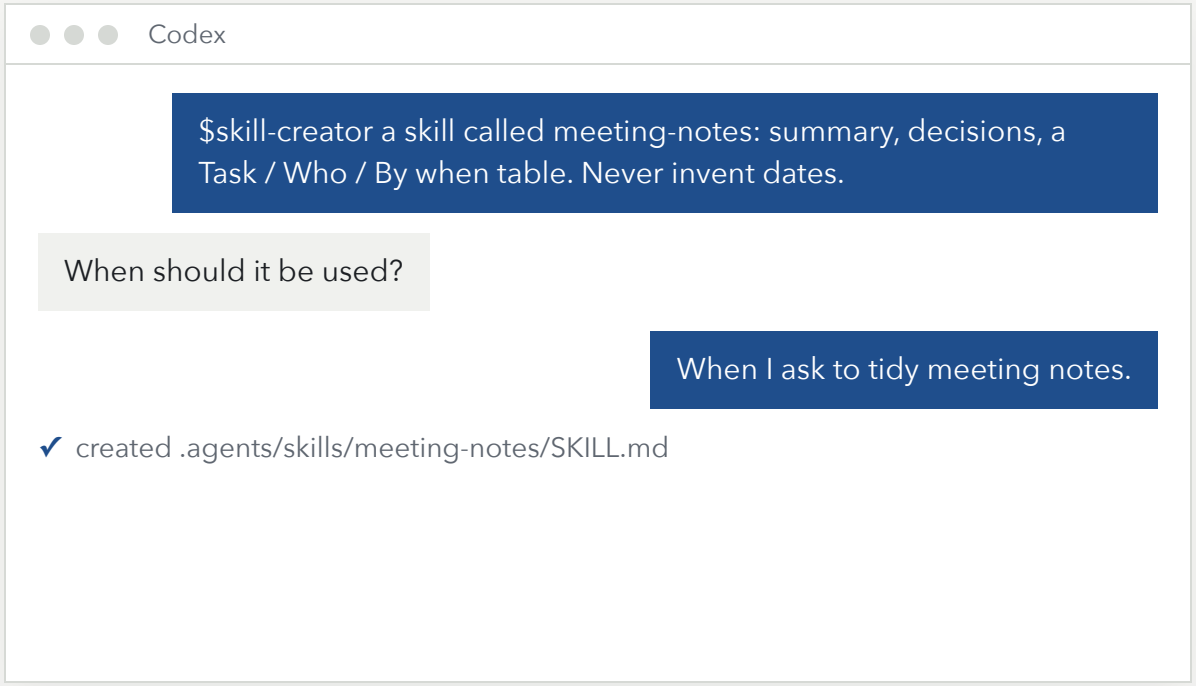


Asks how blanks count; one number to check by hand

Removes names and emails first; exact quotes

One file, works offline, saves as PDF

Make your own skill in one prompt



→

SKILL.md

```
name: meeting-notes
description: Tidy
meeting notes. Not for
other documents.
---
1. Summary
2. Decisions
3. Action table
Mark unclear with [CHECK]
```

Then use it anywhere: **$meeting-notes**. Improve it with one more prompt.

Slides that do not look AI-generated

International visitors rate language support lowest

2.6



editorial

Onsen is the strength for everyone

90%



bold

Buses draw 23 of 83 comments

23



field

✓ Title = the takeaway · one accent · a source on every number

✕ Gradients · emoji bullets · "unlock", "elevate", "delve"

Your turn: pick a route

00 · 01
Install, first task

→

Prompts
Six parts

→

Do you have a project?
Choose

Yes
04A → 02 → 03 → 04B
your data, your skills

No, I like surveys
98 · Survey to slides
skills included

No, I like data
99 · Thesis data
then make a skill

github.com/Pann13223029/codex-seminar

Check one number yourself

```
=COUNTIFS(A:A,2024,H:H,">0")/COUNTIF(A:A,2024)
```

fiscal_year	…	power_capacity_kw	
2024	…	1200	
2024	…		**0.411**
2024	…	0	

1. Recompute by hand
2. Ask how blanks were treated
3. Check every reference exists
4. Disclose your AI use

Undisclosed AI help can count as plagiarism. Check your course rules; template in step 03.

Four ways to get it wrong

✕ Private data in the folder

Interviews, names, passwords: keep them out

✕ Invented references

They look real. Search for every one

✕ Trusting a confident answer

Ask for the source and the file

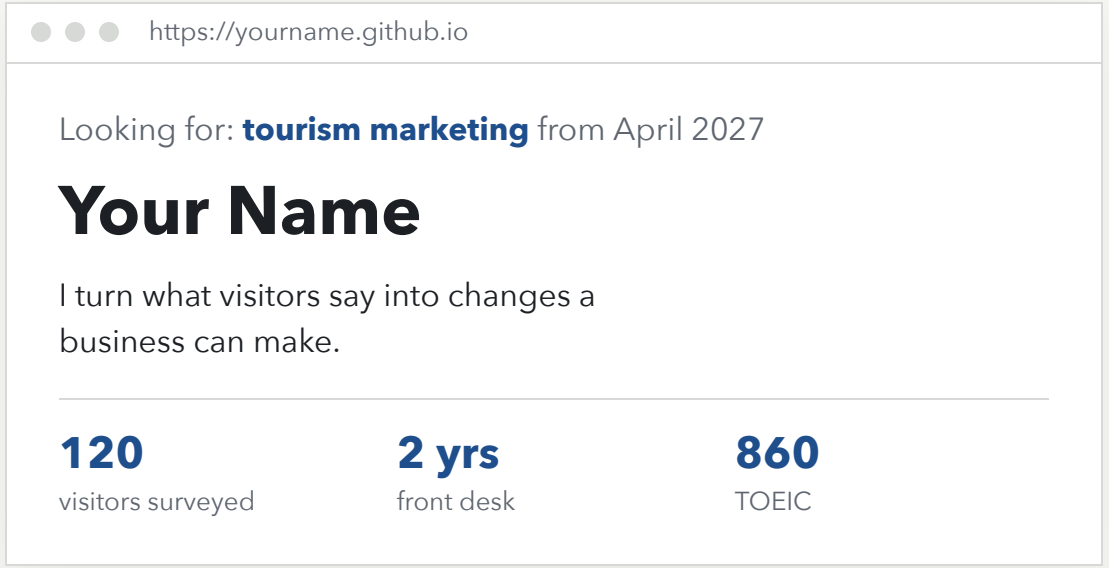
✕ Letting AI do your thinking

The question and the meaning are yours

After class: your portfolio, live

1. **Your CV and notes** into `me/`
2. **$portfolio-page** interviews you
3. **You check** every fact
4. **One page**, English and 日本語
5. **Free on GitHub Pages**

Guide: 97-portfolio-site · about two hours, at home



Example: pannphetra.com · Every number on it must be one you can explain in an interview.

If you remember one thing

Let the agent do the work. Keep the judgement, and check before you trust.

The guide

github.com/Pann13223029/codex-seminar

Questions

Thank you